PBL IDEA

Spatio-Temporal Urban Heat Intensity Prediction Using Machine Learning for Predictive Heat Mapping

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UrbanHeat AI

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Problem:

> Urban areas do not experience the same level of heat. Some areas with dense buildings, roads, and limited vegetation can got higher surface temperatures than areas with more green spaces.

Our idea:

> **UrbanHeat AI** is a mobile-based machine learning system that predicts urban heat intensity by combining spatial, environment, and temporal data, then visualizes the prediction as the interactive heat map.

Background

Urban areas often experience different levels of heat due to differences in vegetation, building density, road coverage, land use, and weather conditions. Some areas with limited vegetation and a higher rate of concentration buildings and paved surfaces can experience higher surface temperatures.

In the other side, identifying potential heat hotspot areas is not just about measuring the real-time temperature. It is also important to understand how location, environment conditions, and time influence heat intensity.

Currently, temperature and environmental data often available from different sources. However, these data can be difficult to combine and interpret into a predictive system that shows where heat intensity potentially be higher.

So from the issue that we raise here, we propose UrbanHeat AI, a machine learning-based system that combines spatial, environmental, and temporal data to predict urban heat intensity and visualize the results through an interactive heat map.

Objectives

The project aims to:

1. Develop a machine learning model to predict urban heat intensity based on spatial, environmental, and temporal factors.
2. Identify location with potentially higher heat intensity through predictive heat mapping.
3. Analyze which environmental factors contribute most to the predicted heat intensity.
4. Provide an accessible mobile interface for visualizing prediction results and heat risk levels.

Project Description

Project Purpose

UrbanHeat AI, mobile apps predictive environmental analysis system that helps users identify potential heat hotspot areas within a selected study area.

Instead of displaying existing temperature information, the system uses **Machine Learning** to learn patterns from historical spatial and temporal data and generate heat intensity predictions.

The prediction results are then visualized through a mobile application developed using **Flutter**.

Proposed Solution

The systems combine several types of data, including:

Spatial Data :

- Geographic coordinates
- Vegetation characteristics
- Land cover
- Building density
- Road/paved surface density
- Elevation, if available

also,

Temporal Data

- Observation date
- Month/season
- Weather conditions
- Other available time-related variables

Target Variable

The primary target variable is planned to be:

Land Surface Temperature (LST)

LST will represent the heat intensity that the machine learning model attempts to predict.

That lean to result of availability and suitability from the dataset that will be validated before implementation.

Project Scope

To keep the project realistic within certain amount of time, the system will focus on a limited study area rather than an entire city or region with the duration of the progress of 12 weeks since this proposal were made..

Scope of Work

The project will include:

1. Gathering relevant spatial data, environmental conditions, and temporal datasets.
2. Preprocessing and integrating data from multiple sources.
3. Creating structured dataset based on location and observation time.
4. Performing exploratory data analysis.
5. Developing and comparing machine learning models.
6. Evaluating model performance using appropriate regression metrics.
7. Applying explainability techniques to identify important prediction factors.
8. Developing a backend/API for serving prediction results.
9. Developing a Flutter mobile application.
10. Visualizing predicted heat intensity through an interactive heat map.

Spatial-Temporal Data Integration

The study area will be divided into several spatial units such as grids or sampling points. Each observation will represent a specific location at a specific time. For example:

Location	Date	Vegetation	Building Density	Road Density	Weather	LST
Grid A	June 1	0.72	0.20	0.10	...	29.4°C
Grid B	June 1	0.35	0.65	0.30	...	32.1°C
Grid A	June 15	0.68	0.22	0.10	...	29.8°C

Machine Learning Approach

Machine Learning Scope

The primary ML task will be formulated as a **regression problem**.

Input

The model will receive environmental, spatial, and temporal features such as:

- Vegetation index/coverage
- Building density
- Road density
- Land cover
- Weather variables
- Location-related features
- Temporal features

Output

The model will predict based on Land Surface Temperature (LST) to convert into **heat intensity** or heat-risk categories for visualization.

Proposed Models

Several machine learning models may be tested and compared, such as:

- Linear Regression as a baseline
- Random Forest Regression
- XGBoost Regression
- Other suitable regression algorithms depending on dataset characteristics

The final model will be selected based on its performance and suitability for the available dataset.

Model Evaluation

The model will be evaluated using regression metrics such as:

- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Square Error)**
- **R^2 (Coefficient of Determination)**

The evaluation will determine how accurately the model predicts heat intensity.

Explainable AI

UrbanHeat AI will not only provide a prediction but will also attempt to explain the factors influencing the prediction.

For example, the system could show that a particular area has a higher predicted heat intensity because of:

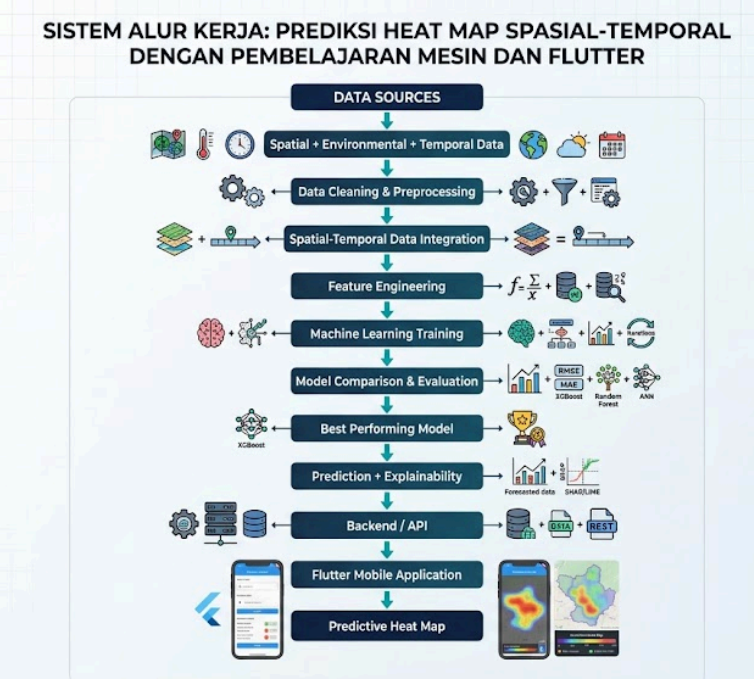
- Low vegetation coverage
- High building density
- High road/paved surface density
- Certain weather conditions

Techniques such as **feature importance or SHAP (SHapley Additive exPlanations)** may used depending on the final model.

This makes the system more interpretable and helps users understand **why an area is predicted to have higher heat intensity**.

System Workflow

The proposed workflow is:



Expected Product

The final product will be a **mobile-based predictive heat mapping application**.

Main Features

1. Heat Map

Displays predicted heat intensity across the selected study area.

Example:

🟢 Low Heat
🟡 Medium Heat
🔴 High Heat

2. Location Detail

Users can click or select a specific area/grid and view:

- Location
- Predicted LST
- Heat level
- Relevant environmental information

3. Prediction

The system provides predicted heat intensity based on the available spatial and temporal conditions.

4. Factor Explanation

The system provides information about the factors that contribute to the prediction.

5. Historical/Temporal View

If sufficient temporal data is available, users can compare heat intensity predictions across different observation periods.

Target Users

The initial target users are:

Primary Users

Urban/environmental planning stakeholders

Potential examples include:

- Local government environmental planners
- Urban planning teams
- Campus/environmental facility management

Secondary Users

- Researchers
- Students
- Environmental communities

The final target user and study area will be adjusted based on the availability of data and potential partner feedback.

Expected Impact

UrbanHeat AI is expected to provide:

Environmental Impact

Help identify areas that may experience higher heat intensity and support awareness of urban heat conditions.

Planning Impact

Provide an additional data-driven reference for prioritizing areas that may require environmental interventions such as vegetation or shading.

Technical Impact

Demonstrate the application of machine learning to a **spatio-temporal environmental prediction problem**.

The system will not directly reduce temperature. Instead, it provides **predictive information that can support future planning and mitigation decisions**.

Limitations

Several limitations are expected in this project:

1. **Data availability**
   The quality and amount of available spatial and temporal data may affect model performance.
2. **Satellite observation limitations**
   Satellite-based observations may not be available for every date and can be affected by cloud cover.
3. **LST limitation**
   Land Surface Temperature represents surface temperature and should not be interpreted as the same as air temperature experienced by humans.
4. **Limited study area**
   Due to the 12-week development period, the project will focus on a selected study area rather than an entire city.
5. **No real-time monitoring**
   The initial system will not use IoT sensors and will not claim real-time temperature monitoring.
6. **Prediction uncertainty**
   Machine learning predictions depend on the quality and representativeness of the training data.

Technology Stack

Component	Technology
Machine Learning	Python
ML Environment	Google Colab
Data Processing	Python / Pandas / NumPy
ML Libraries	Scikit-learn / XGBoost
Explainable AI	SHAP / Feature Importance

Backend	REST API
Mobile Application	Flutter + Dart
Data Visualization	Interactive Map
Version Control	Git & GitHub

Success Criteria

The project will be considered successful if:

1. A structured spatial-temporal dataset can be created from available data sources.
2. At least one machine learning model can successfully predict the selected heat-intensity target.
3. The model can be evaluated using appropriate regression metrics.
4. The system can generate predictions for the selected study area.
5. Prediction results can be visualized as a heat map.
6. The system can provide an explanation of important prediction factors.
7. The machine learning model can be integrated into the Flutter application.
8. The final prototype can demonstrate the complete workflow from input data to heat prediction.

Conclusion

UrbanHeat AI proposes a machine learning-based approach to predict urban heat intensity by combining spatial, environmental, and temporal information.

Unlike a conventional monitoring system, the proposed system focuses on **prediction**, allowing potential heat hotspot areas to be identified based on learned patterns from historical data.

The project will combine **Python-based machine learning, spatial-temporal data processing, explainable AI, API integration, and Flutter mobile development**.

To maintain feasibility within the 12-week PBL period (since this proposal was released), the project will use a **limited study area**, existing/public datasets, and will not require IoT hardware.

The feasibility of the project will first be validated through **dataset availability, quality, and study-area selection** before the implementation phase begins.